Athena's Light - Accessible, highly portable research combining fNIRS, EEG, & PPG

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***Abstract*—Muse headbands are commercially available biosensing devices designed to measure human brain and related biometric data for meditation, research, and neurofeedback training using EEG, PPG, and fNIRS. This study explores the use of Muse devices with MuseCroc Mobile, a research tool currently under development for use in neuroscience research. Topics covered include preliminary validation of the data collection pipeline by observing changes in brain activity during eyes-open and eyes-closed conditions, analyzing the data stream processing behavior of MuseCroc Mobile, and the use of this software for developing an understanding of the fNIRS system of the Muse S Athena.**

I. Introduction

MuseCroc Mobile is a research tool used to connect, control, and record data from Muse headbands [1]. MuseCroc Mobile was developed at MannLab in Toronto, Canada and released as part of the 27th Mersivity Proceedings [2]. MuseCroc Mobile provides tools for researchers to conduct research on human brain activity and understand the function and behavior of the EEG, PPG, and fNIRS systems in Muse devices to improve research quality. This study presents initial investigation into the neurological effects of experimental flows delivered via a phone application, demonstrates how device control can be used to understand the functioning of Muse devices, and provides an analysis of the limitations and behavior of the software.

II. Background

A. Electroencephalography (EEG) & Muse EEG

Electroencephalography (EEG) is a method of measuring electrical activity of the brain, first discovered more than 100 years ago [3]. The Muse is a compact, low-cost EEG headset used in research, including sleep studies [4], assessment of well-being indicators [5], Event Related Potentials (ERP) [6], EEG research on audiences of people [7], and mobile method of assessment of cognitive fatigue [8].

The built-in electrodes in the Muse devices map to TP9, AF7, AF8, and TP10 in the 10-20 system [9], and additional AUX electrodes can be added via the device's USB port and enabled by changing the device settings using the SDK. The Muse headset outputs raw EEG data in µV, and can generate processed relative band powers [1].

B. Functional Near Infrared Spectroscopy (fNIRS)

Functional Near Infrared Spectroscopy (fNIRS) is a technology used to measure Oxyhemoglobin (HbO) and Deoxyhemoglobin (HbR) in the human brain [10], [11]. fNIRS systems operate by measuring near infrared light at multiple wavelengths, such as 730nm and 850nm, as they pass through human brain tissue [10]. Different wavelengths of infrared light have different patterns of absorption, allowing operators to measure changes in blood flow and hemoglobin oxygenation in the brain [10], [11]. Research conducted using fNIRS includes assessment of task-related cortical function and monitoring of cerebral oxygenation in neonatal ICUs [11] and the neural mechanisms of addiction [12].

The Muse S Athena is a compact headband providing fNIRS functionality in addition to the EEG, PPG, & related sensors present in other Muse headbands. The Muse S Athena allows access to raw sensor output for the fNIRS system in µA, however the method of interpretation is currently under investigation [**?**].

C. PPG - Heart Rate Tracking

Photoplethysomography (PPG), is a non-invasive method of measuring changes in blood volume in the microvascular bed of tissue to evaluate the function of the cardiovascular system, including heart rate [13]. Recent Muse headbands provide appropriate hardware for PPG [1].

D. Software & Resource Development

The user interface of the application was migrated to Jetpack Compose from the previous Java base. This enables rapid development of flexible, dynamic UI components compared to previous prototypes. These visualizations can be seen in figure 1 and 2.

Muse headbands can be configured to use different presets that dictate the type of data the device collects. MuseCroc mobile allows for control over the preset used, enabling power consumption and data collection to be optimized together for longer experiments. In addition, the MuseCroc Mobile software has been improved to force reconnection to disconnected headbands, enabling better coverage in dynamic environments or in case of connection issues.

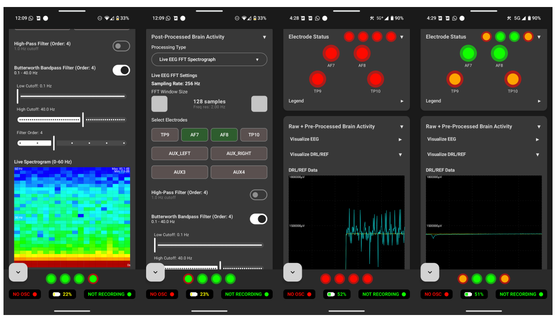


Fig. 1: MuseCroc Mobile - Live Signal Processing & Quality Analysis Screenshots (Prototype Stage)

Improvements were made to the visualization of connection quality and brain activity. Circles are used to represent the built-in electrodes, arranged left to right (TP9, AF7, AF8, TP10). "HSI", or connection quality as indicated by the Muse SDK [1] is represented by the inner circle, while the outer circles represent artifact detection as indicated by the "IS_GOOD" packets given by the Muse SDK [1]. A live rolling windowed fast fourier transform has been added to visualize EEG data for live interpretation of the EEG signal. This visualization includes toggles for selected electrode sources, window size in samples, and adjustable high-pass and bandpass butterworth filters for interference removal. Similar tools have also been provided for graphing raw EEG, DRL/REF sensor activity, raw optics (PPG/fNIRS), Gyroscope/Accelerometer, and relative band powers, as summed or averaged values across selected sources and waves.

Software used for Muse research has historically included Open Sound Control (OSC) streaming as a method of transmitting data [14]. Due to this, OSC streaming has been added to MuseCroc Mobile. OSC was originally intended for music production, and is a lightweight variant of User Datagram Protocol (UDP) using a URL-like addressing system to identify information [15].

The application now includes pre-written experiments. These experiments are presented without distractions on the phone screen, and automatically include experiment specifications and user interactions as part of the generated .csv files. In addition, a trigger button has been added for manually tracking experiment events. Finally, recordings and recording control have been improved to include collected data types and sampling rates in the filename.

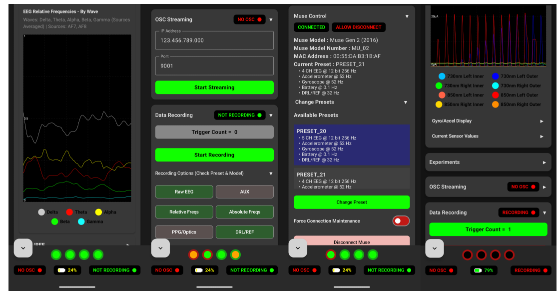


Fig. 2: MuseCroc Mobile - Live Visualization & Device Control Screenshots (Prototype Stage)

E. EEG Activity in an eyes open or closed state

In cognitive neuroscience, a well-established finding is that occipital alpha-band (Roughly 10 Hz) activity becomes stronger when the eyes are closed compared to when they are open [16]. This phenomenon has been interpreted as evidence that alpha oscillations reflect a default dynamic state of the visual system that emerges in the absence of external input. Later studies demonstrated that increased alpha power is accompanied by reduced cortical spiking activity [17], suggesting that elevated alpha oscillations during eyes-closed conditions correspond to a baseline or inhibitory state of visual processing. Together, these findings support the view that alpha-band activity represents the visual brain's intrinsic resting mode, to which it returns when sensory stimulation is absent [18]. In our exploratory analysis comparing eyes-open and eyes-closed conditions, we observed an increase in alpha power over occipital regions, consistent with the previous literature.

F. Motivation

This study aims to assess and demonstrate the usability of MuseCroc Mobile for use in neuroscience research, evaluate the Muse S Athena fNIRS system, and analyze how MuseCroc Mobile processes Bluetooth packets.

The long term goal of this project is to enable accessible neuroscience research by developing this software to a stable, validated state before distribution to the broader public, free of charge.

III. Methodology

A. Open & Closed Eyes Experiment

1) Participants and Data Acquisition: EEG data was recorded from four participants using four scalp electrodes (TP9, AF7, AF8, and TP10) positioned according to the international 10–20 system [19]. Participants were university students in the 17-26 age range, tested in controlled conditions with the experiment flow being delivered by a phone resting on a desk. Signals were sampled at 256 Hz and stored in microvolts (µV). The experimental protocol randomly alternated between eyes-open and eyes-closed resting-state conditions for 20 runs per participant, with each run lasting 15 seconds. Two participants were excluded from analysis due to excessive artifacts and poor electrode connection that rendered their data unreliable for further processing.

2) Data Preprocessing: Raw EEG data were processed using the MNE-Python [20]. Channel names and types were assigned. Voltage values were converted from µV to V. A zero-phase finite impulse response (FIR) band-pass filter from 0.1 Hz to 30 Hz was applied to remove slow drifts and high-frequency noise. Additionally, a 60 Hz notch filter was used to suppress power-line noise. The continuous EEG was then segmented into 15-s epochs (or windows of data) time-locked to each event, producing separate sets of epochs for eyes-open and eyes-closed conditions. Baseline correction was applied at the beginning of each epoch to minimize slow potential shifts [21]. Only epochs that remained free of artifacts or excessive noise, as determined by automated thresholding procedures in MNE-Python, were retained for subsequent analysis [22].

3) Spectral Analysis: Power spectral density (PSD) estimates were computed for each epoch and channel using Welch's method as [23]. PSDs were calculated over the 1–40 Hz frequency range. Human EEG largely comprises signal power in a range from about 1–30 Hz (with some extension to 40 Hz) [24], and values were expressed in (µV²/Hz). For each electrode, PSDs were averaged across epochs to obtain stable mean power spectra for the eyes-open and eyes-closed conditions. Alpha-band power (8–12 Hz) was extracted by averaging PSD values within that frequency range. To quantify condition differences, alpha power during eyes-closed periods was compared to that during eyes-open periods both numerically and topographically. Topographic scalp maps were generated to illustrate the spatial distribution of alpha-band activity using MNE's spherical interpolation method. The eyes-closed condition typically exhibited stronger alpha power in posterior (occipital and parietal) regions, consistent with well-established physiological patterns. Moreover, Alpha-band power was averaged across epochs within each condition and compared between eyes open and closed.

B. Muse S Athena fNIRS System Investigation

The fNIRS system of the Muse S Athena consists of inner and outer channels on both sides of the headband, and is indicated by the Muse SDK as measuring 730 nm wavelength infrared light, 850 nm wavelength infrared light, Ambient light, and Visible Red light, with values recorded in µA [**?**]. Although this paper will not attempt to use the raw fNIRS data generated by this device to analyze brain hemodynamics, we demonstrate how preset control via MuseCroc Mobile can be used to analyze the system and its function by controlling the devices and measuring light emitted from the central optode. These experiments intend to support future investigation and interpretation of Muse S Athena fNIRS data.

C. CSV File Writing & Packet Processing Analysis

It is important to consider that the Android Operating System is not considered qualified for real-time uses [25] and thus may not be able to meet all millisecond or sub-millisecond scale requirements in neuroscience research. Research using older Muse models with certain operating systems has been found to experience jitter, although institutions that rely on Muse devices do not necessarily consider it detrimental [26]. In MuseCroc Mobile, The Foreground Service writes a line to the CSV file for each Bluetooth packet received in sequence. Thus, it is reasonable to use the distance between lines written to the file as a way to observe packet processing and file write behavior of the program. Packets are processed as groups in the order they are received.

IV. Results

A. Open & Closed Eyes Experiment

As shown in Figure 3, there is a clear increase in the power of the alpha band in the occipital regions during the closed eye condition compared to the open eye. Compared with the average alpha power shown in Figure 4 this difference remains more pronounced for the closed eye condition. Consistent with well-established physiological findings, the closed eye state exhibits stronger alpha activity in posterior (occipital and parietal) regions [27].

B. Preliminary fNIRS System Analysis

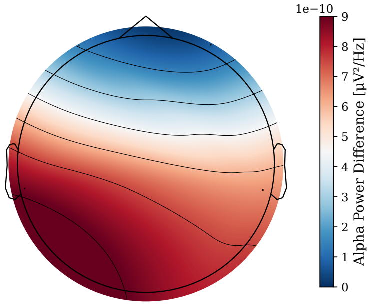
Although the Muse S Athena provides data from the fNIRS system, the exact method of interpretation of these values is currently not publicly available. In order to begin the process of interpreting these values, we have begun investigating the function of the device. Throughout this investigation, measurements of 730 nm and 850 nm Near-Infrared Light were attempted using the Muse S Athena's fNIRS system on High-Power OPTICS mode presets.

1) Muse S Athena Optode Dimensions & Relative Positioning: Optode measurements and descriptions taken from the outside of the device can be seen in table I and II, with bilateral symmetry of both inner and outer channels.

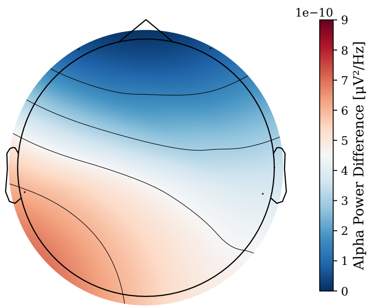
TABLE I: Optical Sensor Configuration and Specifications

Component	Dimensions (mm)	Distance From Center (mm)
Inner Channel	5 × 4	24
Outer Channel	3 × 3	4
Central LED	3 × 3	0
Central Sensor	2 × 3	0

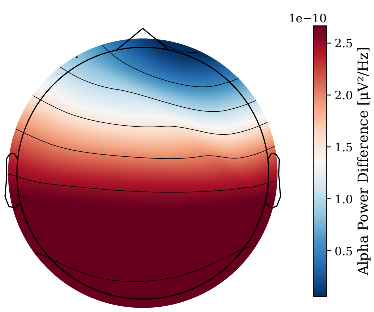
2) Wattage per cm^2 at 1-inch distance intervals: The Muse S Athena was placed on a CNC Machine and backed away from the optical power sensor connected to an optical power meter at 1-inch intervals ranging from 0 to 15 inches. The optical power meter used an attenuator to suppress line noise and ambient readings were taken at the same distances, revealing ambient levels under 100 µW/cm^2. The meter displayed



(a) Eyes-closed



(b) Eyes-open



(c) Difference (closed − open)

Fig. 3: Topographic maps of alpha power (8–12 Hz) distribution across the scalp during (a) eyes-closed and (b) eyes-open conditions, with (c) showing the difference map (eyes-closed minus eyes-open).

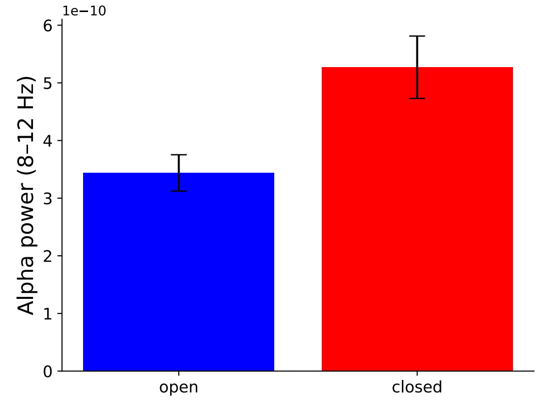


Fig. 4: Average alpha power (8–12 Hz) during eyes-open and eyes-closed conditions. Error bars represent the standard error of the mean (SEM).

TABLE II: Optode Component Descriptions

Component	Description
Inner Channel	Opaque rectangular sensor with transparent membrane
Outer Channel	Opaque square sensor in circular transparent membrane
Central LED	Light emitting diode
Central Sensor	Opaque sensor above LED

an overload indicator until 7 inches distance for 730 nm light, and 9 inches for 850 nm light. Minimum and maximum values displayed by the meter were visually observed and recorded, seen in figure 5. This method is limited by human error, since exact maximum and minimum measurements may be difficult to identify in a fluctuating output.

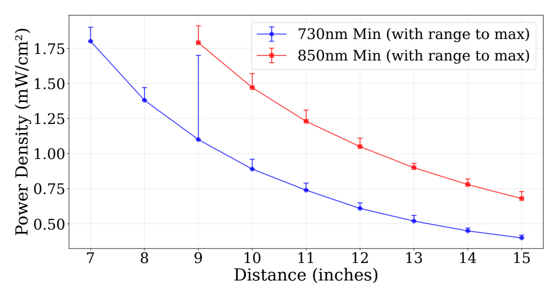


Fig. 5: Minimum Optical power measurements by distance for 730 nm and 850 nm wavelengths

3) Amperage output by a Muse S Athena's left inner channel optode: When the outer channel optode on a Muse S Athena is exposed to the light of a separate Muse S Athena, an interesting pattern emerges. Rather than providing a flat, distance-dependent pattern of visualization as initially expected, an arching pattern separated by low, flat areas becomes apparent in the live raw fNIRS graph.

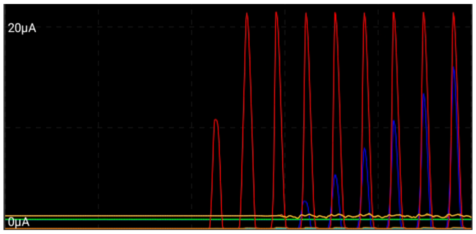


Fig. 6: Live raw optics graph from MuseCroc Mobile during a dual-muse experiment

To investigate the relationship between light exposure of the inner optode and output in μA, we observed the highest recorded values per distance. These values are graphed in Figure 7.

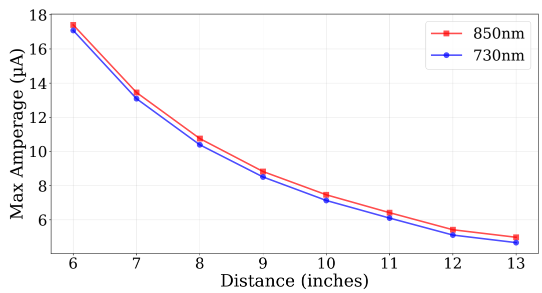


Fig. 7: Measured maximum recorded and distance.

4) Oscilloscope analysis of voltage output of the optical sensor exposed to the central optode: In order to further investigate the observed pattern and behavior of the amperage and wattage readings, the optical sensor was removed from the optical power meter and connected to an oscilloscope while pressed to the Athena's central optode. Analysis of the signal provided a frequency of 64.265 Hz, taking the shape of the waveform visible in Figure 8. The observed frequency of the signal is roughly the same as the fNIRS sampling rate of the Athena at 64 Hz.

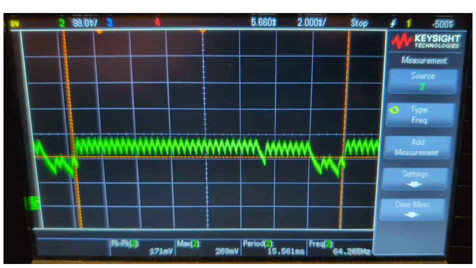


Fig. 8: Oscilloscope showing the signal emitted via the central optode

5) Preliminary Investigation of Muse S Athena optical emission spread: A preliminary investigation of emission pattern was performed by placing the optical power meter 10 cm away from the central emitting optode of a Muse S Athena resting horizontally on a hand-drawn protractor, with results recorded in Figure 9. However, this method did not account for the optical properties of the protractor material, and the error tolerance of the position is unknown.

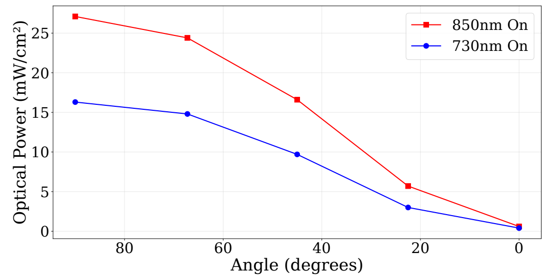


Fig. 9: Optical Power by angle, 10cm Distance

C. File Write & Packet Processing Analysis

TABLE III: EEG Recording Statistics for Eyes Open/Closed Experiments

Session	Duration (ms)	Actual Samples	Expected Samples	Sample Diff.	Avg. IST (ms)
0001	350,117	89,840	89,630	+210	3.897
0002	379,117	97,308	97,054	+254	3.896
0003	388,865	99,852	99,549	+303	3.894
0004	356,504	91,552	91,265	+287	3.894

IST = Inter-Sample Time; Expected IST at 256 Hz: 3.906 ms

Looking at table III, we see that there are between 210 and 303 'extra' packets for each session compared to the expected count.

TABLE IV: EEG Packet Inter-Arrival Time Statistics

Session	Inter-Packet Time (ms)					
	Mean	Std Dev	75%	95%	99%	Max
0001	3.897	11.422	1.000	21.000	69.000	139.000
0002	3.895	11.817	1.000	21.000	69.000	141.000
0003	3.894	11.304	1.000	20.000	68.000	137.000
0004	3.893	11.379	1.000	21.000	68.000	127.000

Observing Table IV, we see that at least 75% of packets are recorded as processed and written to file within 1 ms or less of each other, while the mean inter-packet time is roughly 3.89 ms, with an expected inter-packet write time of 3.906 ms.

Uneven packet spacing is common in studies involving Muse headbands connected via Bluetooth, however this is not necessarily considered cause for concern [26]. In addition, Android is not considered suitable for real-time uses [25], and this limits our ability to provide truly exact parameters of program function. Research with Muse devices can be conducted using this software, although it is recommended to observe packet processing behavior to ensure adequate transmission and processing. The expected behavior of this program is to queue the packets until they are processed as a group, resulting in uneven write patterns.

V. Conclusion & Further Research

The eyes open & closed experiment provides initial validation of the pipeline formed by the MuseCroc Mobile software in combination with the Muse S Athena. In our exploratory analysis comparing eyes-open and eyes-closed conditions, we observed an increase in alpha power over occipital regions, consistent with the previous literature. This result confirms that our recording setup effectively captures phase-related variations in neural activity and reliably reproduces established physiological responses. However, further investigation is needed to establish reliability and improve data collection procedures. The live visualizations and preset control provided by the application have proved useful for analyzing the behavior of the Muse S Athena's Optics system, providing a tool that can be used in further research of the hardware that enables fNIRS. Packet processing and line writing analysis have provided preliminary validation of the software for data collection, and will be used to inform future research and development.

VI. Acknowledgment

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References

[1] Mitchell Seitz, Nishant Kumar, Homey Poon, Rocklen Jeong, Aydin Hosseingholizadeh, and Steve Mann. Musecroc mobile - highly accessible eeg, ppg, and fnirs data collection visualization, published in the 27th annual mersivity / water-hci symposium, 2025.

[2] Steve Mann, Michael Condry, and Nishant Kumar, editors. *27th Annual Mersivity/Water-HCI Symposium*. Zenodo, August 2025. DOI: 10.5281/zenodo.16973160.

[3] Paolo M. Rossini, Jonathan Cole, Walter Paulus, Ulf Ziemann, and Robert Chen. 1924–2024: First centennial of eeg. *Clinical Neurophysiology*, 170:132–135, 2025.

[4] Steve Mann, Nishant Kumar, Joao Pedro Bicalho, Malek Sibai, and Calum Leaver-Preyra. Adaptive chirplet transform-based sleep state detection. In *2025 IEEE International Conference on Consumer Electronics (ICCE)*, pages 1–6, 2025.

[5] Cédric Cannard, Arnaud Delorme, and Helané Wahbeh. Identifying hrv and eeg correlates of well-being using ultra-short, portable, and low-cost measurements. *bioRxiv*, 2024.

[6] Olave E. Krigolson, Chad C. Williams, Angela Norton, Cameron D. Hassall, and Francisco L. Colino. Choosing muse: Validation of a low-cost, portable eeg system for erp research. *Frontiers in Neuroscience*, Volume 11 - 2017, 2017.

[7] Georgios Michalareas, Ismat M.A. Rudwan, Claudia Lehr, Paolo Gessini, Alessandro Tavano, and Matthias Grabenhorst. A scalable and robust system for audience eeg recordings. *bioRxiv*, 2022.

[8] Olave E. Krigolson, Mathew R. Hammerstrom, Wande Abimbola, Robert Trska, Bruce W. Wright, Kent G. Hecker, and Gordon Binsted. Using muse: Rapid mobile assessment of brain performance. *Frontiers in Neuroscience*, Volume 15 - 2021, 2021.

[9] Aydin Hosseingholizadeh, Nishant Kumar, and Steve Mann. Portable eeg-based data acquisition and multi-sensor integration using the muse s headband. In *2025 International Conference on Control, Automation and Diagnosis (ICCAD)*, pages 26 – 32, 2025. DOI: 10.5281/zenodo.16973160.

[10] Kaleb T. Kinder, Hollis L. R. Heim, Jessica Parker, Kara Lowery, Alexis McCraw, Rachel N. Eddings, Jessica Defenderfer, Jacqueline Sullivan, and Aaron T. Buss. Systematic review of fNIRS studies reveals inconsistent chromophore data reporting practices. *Neurophotonics*, 9(4):040601, 2022.

[11] Wei-Liang Chen, Julie Wagner, Nicholas Heugel, Jeffrey Sugar, Yu-Wen Lee, Lisa Conant, Marsha Malloy, Joseph Heffernan, Brendan Quirk, Anthony Zinos, Scott A. Beardsley, Robert Prost, and Harry T. Whelan. Functional near-infrared spectroscopy and its clinical application in the field of neuroscience: Advances and future directions. *Frontiers in Neuroscience*, Volume 14 - 2020, 2020.

[12] Alessandro Carollo, Ilaria Cataldo, Seraphina Fong, Ornella Corazza, and Gianluca Esposito. Unfolding the real-time neural mechanisms in addiction: Functional near-infrared spectroscopy (fnirs) as a resourceful tool for research and clinical practice. *Addiction Neuroscience*, 4:100048, 2022.

[13] John Allen. Photoplethysmography and its application in clinical physiological measurement. *Physiological Measurement*, 28(3):R1, feb 2007.

[14] Olave E Krigolson, Chad C Williams, Angela Norton, Cameron Dale Hassall, and Francisco L Colino. Choosing muse: Validation of a low-cost, portable eeg system for erp research. *Frontiers in Neuroscience*, 11:109, 2017.

[15] Matthew Wright and Adrian Freed. Open sound control: A new protocol for communicating with sound synthesizers. In *Proceedings of the International Computer Music Conference (ICMC)*, pages 101–104, Thessaloniki, Greece, 1997.

[16] Hans Berger. Über das Elektrenkephalogramm des Menschen. *Archiv für Psychiatrie und Nervenkrankheiten*, 87:527–570, 1929.

[17] Saskia Haegens, Vahé Nácher, Román Luna, Ranulfo Romo, and Ole Jensen. α-Oscillations in the monkey sensorimotor network influence discrimination performance by rhythmical inhibition of neuronal spiking. *Proceedings of the National Academy of Sciences*, 108(48):19377–19382, 2011.

[18] G. Pfurtscheller, A. Stančák, and C. Neuper. Event-related synchronization (ERS) in the alpha band—an electrophysiological correlate of cortical idling: A review. *International Journal of Psychophysiology*, 24(1–2):39–46, November 1996.

[19] G. H. Klem, H. O. Lüders, H. H. Jasper, and C. E. Elger. The ten-twenty electrode system of the International Federation. The International Federation of Clinical Neurophysiology. *Electroencephalography and Clinical Neurophysiology. Supplement*, 52:3–6, 1999.

[20] Alexandre Gramfort, Martin Luessi, Eric Larson, Denis A. Engemann, Daniel Strohmeier, Christian Brodbeck, Roman Goj, Mainak Jas, Teon Brooks, Lauri Parkkonen, and Matti Hämäläinen. Meg and eeg data analysis with mne-python. *Frontiers in Neuroscience*, Volume 7 - 2013, 2013.

[21] Burkhard Maess, Erich Schröger, and Andreas Widmann. High-pass filters and baseline correction in M/EEG analysis. Commentary on: "How inappropriate high-pass filters can produce artefacts and incorrect conclusions in ERP studies of language and cognition". *Journal of Neuroscience Methods*, 266:164–165, 2016.

[22] Alexandre Gramfort, Martin Luessi, Eric Larson, Denis A. Engemann, Daniel Strohmeier, Christian Brodbeck, Lauri Parkkonen, and Matti S. Hämäläinen. MNE software for processing MEG and EEG data. *NeuroImage*, 86:446–460, 2014.

[23] Peter D. Welch. The use of fast Fourier transform for the estimation of power spectra: A method based on time averaging over short, modified periodograms. *IEEE Transactions on Audio and Electroacoustics*, 15(2):70–73, 1967.

[24] J. Lanzone, M. A. Colombo, S. Sarasso, F. Zappasodi, M. Rosanova, M. Massimini, V. Di Lazzaro, and G. Assenza. EEG spectral exponent as a synthetic index for the longitudinal assessment of stroke recovery. *Clinical Neurophysiology*, 137:92–101, May 2022.

[25] Luc Perneel, Hasan Fayyad-Kazan, and Martin Timmerman. Can android be used for real-time purposes? In *2012 International Conference on Computer Systems and Industrial Informatics*, pages 1–6, 2012.

[26] Olav E. Krigolson. MUSE data collection: Guidelines and protocols, n.d. Laboratory documentation for EEG data collection using the MUSE headband. Accessed: 2025-10-30.

[27] D. P. X. Kan, P. E. Croarkin, C. K. Phang, and P. F. Lee. EEG differences between eyes-closed and eyes-open conditions at the resting stage for euthymic participants. *Neurophysiology*, 49(6):432–440, December 2017.